

Emmanuel Acheampong Oppong

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EDUCATION

Grambling State University

Expected May 2028

Bachelor of Science in Computer Science, 3.95/4.0 GPA

- **Relevant Coursework:** Object-Oriented Programming, Database Management, **Data Structures & Algorithms**, Operating Systems, Computer Networks, Cloud Computing

TECHNICAL SKILLS

Programming Languages: Python, Java, JavaScript, C++, HTML/CSS, SQL

Frameworks & Libraries: React, Node.js, Spring Boot, Angular, Django, Bootstrap

Tools & Platforms: Git, Docker, Kubernetes, AWS, Jenkins, JIRA

Databases: MySQL, MongoDB, PostgreSQL, Oracle, Redis, SQLite

EXPERIENCE

Google Basta Fellow (GSWEP), Google

Aug 2026 – Present

- Developed a **Python** and **TensorFlow** classification model to identify anomalous records in historical operational data, optimizing the preprocessing and inference pipeline to reduce data-processing time by **35%**
- Designed and implemented an **automated testing framework** for a **Python**-based data-processing platform, integrating unit and regression tests into **CI** and reducing bug occurrence by **40%** across a five-engineer team
- Redesigned the **React** interface for an internal batch-monitoring tool used by **40+ engineers**, simplifying job-status workflows and increasing user-experience survey scores by **50%**
- Optimized **20+ PostgreSQL** queries supporting the batch-monitoring tool by adding indexes and restructuring high-cost joins, reducing p95 query latency by **25%**

Software Engineer Fellow, HeadStarter AI

Jul 2026 – Oct 2026

- Developed a **Java**-based **AWS** resource-allocation tool automating **EC2** and **S3** provisioning, saving about **10 hours** of manual work per month
- Implemented **AES-256 encryption** for sensitive data at rest and integrated **AWS KMS** for managed key storage, reducing high-severity security findings by **25%** in follow-up vulnerability assessments
- Reviewed **150+ Java** and cloud-automation pull requests for correctness, security, and maintainability, cutting average review turnaround time by **30%**

Bloomberg Tech Lab Fellow, Bloomberg

Oct 2025 – Dec 2025

- Designed and deployed **Spring Boot** microservices on **Kubernetes**, enabling independent scaling of high-traffic services and increasing supported request throughput by **35%** in load tests
- Integrated **Bloomberg Open API (BLPAPI)** to stream real-time prices and historical data for **10+ financial instruments**, powering live portfolio monitoring
- Analyzed **PostgreSQL** execution plans and optimized indexes and high-cost joins, reducing p95 database latency by **30%** for the market-data service
- Built **Jenkins**, **Docker**, and **Kubernetes CI/CD** pipelines for automated testing, image publishing, and rolling deployments, reducing manual release effort by **40%**

IT Assistant, Grambling State University

Sep 2024 – Apr 2025

- Deployed software updates and security patches across **500+ computers** in **60+ campus buildings**, achieving **100% update completion** with zero reported disruptions
- Configured and maintained switches, wireless access points, and network connections across **60+ campus buildings**, sustaining **95% network uptime** campus-wide
- Built and maintained an **Excel**-based inventory system tracking **500+ IT assets** across **60+ campus buildings**, reducing average asset lookup time by **40%** and speeding up audit completion

PROJECTS

Stock Price Prediction (AI4ALL) | Python, FINBert, Scikit-learn

- Built an end-to-end **financial NLP pipeline** processing **322K headlines** and **20M stock records** using **FINBert** sentiment analysis for stock-movement classification
- Engineered a novel Competitive Pressure Index capturing negative news surges relative to each stock's baseline coverage, serving as the core feature for downstream classification
- Trained a **Random Forest classifier** via 5-fold cross-validation achieving **54%** accuracy, outperforming a **Logistic Regression** baseline with full confusion matrix evaluation

CITIES Dashboard | React, Material UI, Google Charts, GitHub Actions

- Built the front-end of a dashboard delivering air quality insights to **20K+ students** across Monroe, Louisiana schools
- Automated deployment workflows using **GitHub Actions**, replacing manual build-and-release steps with a single pipeline and cutting release time by roughly **30%**

LEADERSHIP & PROFESSIONAL DEVELOPMENT

ColorStack Grambling Academic Chair

Aug 2026 – Present

Grambling State University

- Led weekly **LeetCode** workshops for **90+ members** on data structures and algorithms, helping **15+ members** land **software engineering internships**

Other Contributions: JPMorgan Data for Good Hackathon (April 2026), Open-Source contributions: **Apache Kafka**, **pgjdbc**, **kubernetes-client/java**, **facebook/react** (**20+ PRs**), NSBE Treasurer, Lantern Network Ambassador